

1 Bias and Power of Slope Versus Final-Visit Estimands Under
2 Nonlinear Treatment Effects in Alzheimer’s Disease Trials

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32 **Abstract**

33 We use Monte Carlo simulation to quantify how nonlinearity in the treatment effect tra-
 34 jectory affects the bias, coverage, and power of a random-slopes mixed model relative to a
 35 categorical-time MMRM, in the setting of an 18-month Alzheimer’s disease trial with CDR-
 36 SB as the endpoint and a fixed total treatment effect of -0.45 points at month 18 (matching
 37 CLARITY-AD). Across seven values of a shape parameter κ spanning early-onset to delayed-
 38 onset treatment effect trajectories, the random-slopes model is approximately unbiased only
 39 near $\kappa = 1$ (exact linearity) and its bias grows to 0.227 points (versus a true effect of -0.45)
 40 at the most convex trajectory examined ($\kappa = 0.3$); its 95% CI coverage falls as low as 0.815
 41 in the same condition. The categorical-time MMRM remains approximately unbiased and
 42 maintains coverage at or above 0.936 across all conditions examined. Contrary to the con-
 43 ventional expectation that a correctly specified parsimonious mean structure is more efficient,
 44 the random-slopes model does not achieve higher power than the categorical-time MMRM
 45 even under exact linearity in this design; we show this is attributable in part to an estimator
 46 choice (an unidentified treatment main effect retained in a model fit to post-baseline data
 47 only) rather than to the linearity assumption per se, and provide a preliminary, small-scale
 48 check of a constrained alternative that removes this main effect.

49 **Keywords:** mixed model for repeated measures; random slopes; Alzheimer’s disease; CDR-
 50 SB; estimand; simulation study; nonlinearity.

51 **Data and code availability**

52 The simulation driver (`analysis/scripts/sim_mmr_linear.R`), the report source
 53 (`analysis/report/report.Rmd`), and this compendium’s `renv.lock` are version-controlled
 54 in the `mmrmrobust` repository. `analysis/scripts/run_sim_08.R` reproduces the cached
 55 simulation output loaded by this report (`analysis/data/derived_data/sim_08.rds`); see
 56 the Reproducibility statement at the end of this document for the exact command and RNG
 57 settings used.

58 **1 Introduction**59 **1.1 The CDR-SB as a primary endpoint in AD trials**

60 The Clinical Dementia Rating Sum of Boxes (CDR-SB) has become the primary efficacy
 61 endpoint for disease-modifying trials in early Alzheimer’s disease (AD). The instrument in-
 62 tegrates cognition and function into a single composite score ranging from 0 to 18, and it
 63 demonstrates superior responsiveness to change in early-stage populations compared with tra-
 64 ditional cognitive scales such as the ADAS-Cog^{1,2}. A parallel line of work has instead built
 65 composite cognitive endpoints for the preclinical stage, such as the Preclinical Alzheimer Cog-
 66 nitive Composite³; CDR-SB remains the dominant choice once clinical impairment is present,

67 which is the setting considered here. Annual CDR-SB progression rates in placebo arms of
68 recent trials are approximately 1.3 to 1.7 points per year for participants with baseline CDR
69 global scores of 0.5 to 1.0^{1,4}.

70 1.2 Treatment effects from recent anti-amyloid trials

71 The approval of lecanemab and the positive results for donanemab have provided the first evi-
72 dence that amyloid-targeting therapies can slow clinical decline on CDR-SB. In the CLARITY-
73 AD trial, lecanemab reduced 18-month CDR-SB worsening by 0.45 points (27% slowing) rel-
74 ative to placebo, with the treatment arm declining 1.21 points versus 1.66 in placebo⁵. The
75 TRAILBLAZER-ALZ 2 trial of donanemab demonstrated a similar magnitude of slowing on
76 CDR-SB in participants with low/medium tau burden⁶. By contrast, the earlier Phase 3 stud-
77 ies of aducanumab did not show a consistent CDR-SB benefit across trials⁷, underscoring that
78 both the magnitude and the shape of the treatment-effect trajectory are empirical questions
79 rather than assured features of amyloid-targeting therapy. In the two positive trials, inspec-
80 tion of the CDR-SB trajectories over time reveals a non-constant treatment difference: the
81 separation between arms widens gradually, with the rate of separation appearing to change
82 across the 18-month follow-up period.

83 1.3 The linearity question

84 Both regulatory agencies and the statistical community have debated the choice between two
85 estimands for these trials:

- 86 • **Final-visit estimand:** The between-group difference in mean change from baseline
87 at the last scheduled visit (e.g., month 18). This corresponds to the categorical-time
88 MMRM, which places no parametric constraint on how the mean response evolves over
89 time⁸.
- 90 • **Slope estimand:** The between-group difference in the rate of change per unit time.
91 This corresponds to a random-slopes mixed model that assumes the treatment effect
92 accumulates linearly over the follow-up period.

93 The U.S. Food and Drug Administration’s 2018 draft guidance on early AD drug development
94 discusses the use of slope analyses as a potentially more sensitive approach for detecting
95 disease-modifying effects, particularly when the mechanism is expected to produce a constant
96 rate of slowing⁹. The European Medicines Agency’s guideline similarly acknowledges slope-
97 based endpoints for disease-modification claims, while also noting that the assumption of
98 constant treatment effect rates must be verified¹⁰. The ICH E9(R1) addendum on estimands
99 provides the formal framework for distinguishing these two targets of estimation, emphasizing
100 that the choice of estimand should precede the choice of statistical model¹¹.

101 1.4 Why linearity may not hold

102 Several biological and design factors suggest that the treatment effect in AD trials is unlikely
103 to be perfectly linear in time:

- 104 1. **Delayed pharmacodynamic onset.** Anti-amyloid antibodies require weeks to months
105 to achieve meaningful amyloid clearance. The clinical benefit may lag behind the biolog-
106 ical effect, producing a delayed-onset or concave treatment effect trajectory (the effect
107 accelerates over time).
- 108 2. **Plateau or diminishing returns.** Conversely, if the treatment primarily benefits
109 patients at earlier disease stages, the marginal benefit per additional month of treatment
110 could decrease as disease progresses, producing a convex (early-onset) trajectory.
- 111 3. **Nonlinear placebo progression.** Even if the treatment modifies the underlying rate
112 by a constant fraction, nonlinearity in the natural disease trajectory (documented by¹²
113 using nonlinear mixed-effect models) would produce a nonlinear absolute treatment
114 difference over time.

115 Despite the clinical relevance of this question, limited simulation evidence exists quantifying
116 how much nonlinearity the random-slopes model can tolerate.¹³ compared the slope model
117 with the categorical-time MMRM for ADAS-Cog data, finding that the slope model is more
118 powerful when linearity approximately holds, but biased otherwise. That study, however, did
119 not systematically vary the degree of nonlinearity, nor did it focus on CDR-SB.

120 1.5 Present study

121 We use Monte Carlo simulation to evaluate the robustness of a random-slopes mixed model (a
122 linear mixed-effects model, not itself an MMRM in the clinical-trials sense; see Methods) to
123 departures from linearity in the treatment effect trajectory. We simulate 18-month AD trials
124 with CDR-SB as the primary endpoint, visits at 3-month intervals, and a total treatment
125 effect of -0.45 points at month 18 (matching CLARITY-AD). We parameterize the shape of
126 the treatment effect trajectory using a power function $g(t) = (t/t_{\max})^\kappa$, where $\kappa = 1$ yields a
127 linear effect, $\kappa < 1$ yields an early-onset (convex) pattern, and $\kappa > 1$ yields a delayed-onset
128 (concave) pattern. We compare two analysis models:

- 129 1. A **random-slopes model** that fits treatment-by-time as a linear interaction and ex-
130 tracts the treatment contrast at 18 months.
- 131 2. A **categorical-time MMRM** that fits treatment-by-visit as an unstructured interac-
132 tion and extracts the treatment contrast at the final visit.

133 We evaluate bias, empirical standard error, model-based standard error, power (proportion of
134 $p < 0.05$), and 95% confidence interval coverage across a range of κ values.

135 2 Methods

136 2.1 ADEMP structure

137 Following Morris, White, and Crowther¹⁴, the simulation is reported under the ADEMP
138 framework.

- 139 • **Aims.** Quantify the robustness of a random-slopes model fit (which imposes linear-
140 in-time treatment effects) to nonlinear treatment trajectories governed by the shape

parameter κ .

- **Data-generating mechanism.** A longitudinal model where the treatment effect accrues as $g(t, \kappa) = (t/t_{\max})^\kappa$, with κ varied across $\{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$.
- **Estimand.** The total treatment effect at month 18, $\delta = -0.45$, which is invariant to κ by construction.
- **Methods.** (i) Random-slopes model via `nlme::lme`, which assumes a linear-in-time treatment effect; (ii) categorical-time MMRM via the `mmrm` package with unstructured covariance.
- **Performance measures.** Convergence rate, bias, empirical SE, mean model-based SE, power at $\alpha = 0.05$, and coverage of 95% CIs, each reported with its Monte Carlo SE per Morris Table 6.

2.2 Data-generating model

We generate data for $N = 200$ participants per arm (total $N = 400$) observed at months $t \in \{0, 3, 6, 9, 12, 15, 18\}$. The outcome for participant i at visit j is:

$$Y_{ij} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})t_j^* + \delta \cdot g(t_j, \kappa) \cdot \text{trt}_i + \varepsilon_{ij}$$

where $t_j^* = t_j/12$ is time in years, trt_i is a binary treatment indicator, and:

$$g(t_j, \kappa) = \left(\frac{t_j}{t_{\max}} \right)^\kappa$$

The random effects (b_{0i}, b_{1i}) follow a bivariate normal distribution with standard deviations $\sigma_{b_0} = 1.0$ and $\sigma_{b_1} = 0.6$ (in CDR-SB points per year) and correlation $\rho = 0.3$. The within-subject errors ε_{ij} have an AR(1) structure with marginal variance $\sigma^2 = 1.44$ and autocorrelation $\phi = 0.4$.

2.3 Parameter values

The fixed-effect parameters are calibrated to approximate the placebo arm of CLARITY-AD:

Parameter	Value	Interpretation
β_0	3.2	Mean baseline CDR-SB
β_1	1.5 points/year	Placebo annual decline
δ	-0.45	Treatment effect at $t_{\max}$
κ	Varies	Shape of treatment effect

The κ parameter takes values in $\{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$, spanning strongly convex (early-onset) to strongly concave (delayed-onset) treatment effect trajectories.

2.4 Analysis models

Both models are fit to post-baseline observations only (months 3 through 18); neither model adjusts for baseline CDR-SB, which departs from the near-universal practice of baseline-adjusted change-from-baseline MMRM⁸. We disclose this as a design choice made to keep the two models on comparable footing (a categorical model with visit 3 as the reference level already absorbs baseline-like information from the first post-randomization visit); it is not evaluated as a sensitivity condition here and is noted as a limitation below.

Random-slopes model (unconstrained). We fit a linear mixed-effects model (via `nlme::lme`) with fixed effects for treatment, time (continuous, in years), and their interaction, plus random intercepts and random slopes for time. Because the fixed-effect specification includes a treatment main effect while the model is fit to post-baseline data only (excluding $t = 0$, where the true treatment effect is 0 by randomization), the main effect is not anchored to an observed baseline difference. The treatment effect at 18 months is extracted as $\hat{\beta}_{\text{trt} \times \text{time}} \times 1.5$ only, discarding the fitted main effect; this is an incomplete model-implied contrast rather than the full linear combination $\hat{\beta}_{\text{trt}} + \hat{\beta}_{\text{trt} \times \text{time}} \times 1.5$. As shown in the Results and discussed further below, this estimator choice, not nonlinearity per se, is a plausible driver of the model's power deficit at $\kappa = 1$.

Random-slopes model (constrained). As a remediation, we also implement a constrained variant that omits the treatment main effect entirely ($\mathbf{y} \sim \text{time} + \text{trt}:\text{time}$, random intercept and slope by subject), imposing equality of the two arms at baseline as in constrained longitudinal data analysis (cLDA)^{15,16}. This is standard practice for slope estimands in the trials literature and removes the unidentified main effect implicated above. We report only a small-scale check of this estimator (Results, "Estimator definition check") because a full-scale rerun across all seven κ values and 1000 replicates was outside the time budget of this revision; see the Reproducibility statement for the exact command to complete it.

Categorical-time MMRM. We fit the standard marginal MMRM used in clinical trials (via `mmrm::mmrm`) with fixed effects for treatment, visit (categorical), and their interaction, and a fully unstructured covariance matrix (heterogeneous variances and unstructured correlations across visits). This is the model recommended by⁸, in the general treatment of longitudinal data analysis by¹⁷ and¹⁸, and implemented in the `mmrm` R package¹⁹. The treatment effect at month 18 is estimated as the linear combination of the main treatment effect and the treatment-by-month-18 interaction; its standard error and p -value are obtained from a Satterthwaite-adjusted one-degree-of-freedom contrast (`mmrm::df_1d`), not the delta method.

2.5 Simulation procedure

For each value of κ , we generate 1000 replicate trials, fit both models to each, and record the point estimate, standard error, and p -value for the 18-month treatment contrast. Summary metrics are computed across replicates. For power MCSE ≤ 1.1 pp at $p = 0.5$, $n_{\text{rep}} \geq 2,066$; $n_{\text{rep}} = 1000$ gives power MCSE ≈ 1.58 pp, adequate for initial iteration but should be raised for publication. $N = 200$ per arm is a working design point rather than a value calibrated to a target power for this specific comparison; power calculations for AD trials more generally are

discussed by²⁰. CLARITY-AD itself randomized approximately 900 participants per arm⁵, so the power estimates below should not be read as representative of that trial’s operating characteristics (see Limitations).

3 Results

3.1 Headline findings

We observe that under linear treatment accrual ($\kappa = 1$), both the random-slopes model and the categorical-time MMRM recover the true effect with negligible bias (0.002 and 0.003 respectively, MCSE 0.007 and 0.006). Away from linearity, however, the random-slopes model’s bias grows sharply and its power and coverage are worse than the categorical-time MMRM’s at every κ examined, including $\kappa = 1$: at $\kappa = 1$ the random-slopes model’s power is 0.542 versus 0.623 for the categorical-time MMRM, and its empirical SE (0.221) exceeds the categorical-time MMRM’s (0.191). This contradicts the usual expectation that a correctly specified, more parsimonious mean structure is more efficient, and is addressed directly in the “Estimator definition check” subsection below and in the Discussion. Across all 7 values of $\kappa \in \{0.3, 0.5, 0.7, 1, 1.5, 2, 3\}$, the random-slopes estimator’s largest absolute bias was 0.227 at $\kappa = 0.3$, while the categorical-time MMRM never exceeded 0.007. Coverage of the random-slopes confidence interval ranged from 0.815 to 0.946; the categorical-time MMRM maintained coverage in $[0.936, 0.959]$ across the same range. Power for the random-slopes model ranged from 0.158 to 0.655, and for the categorical-time MMRM from 0.611 to 0.655; the categorical-time model’s power is higher, or effectively tied, at every κ . Full Morris Table 6 MCSEs and convergence rates are reported in Table 1.

3.2 Treatment effect shape

Figure 1 illustrates the treatment effect trajectories $g(t, \kappa)$ for the seven values of κ used in the simulation. When $\kappa = 1$, the treatment effect accrues linearly. When $\kappa < 1$ (e.g., $\kappa = 0.3$), most of the effect is realized early, with diminishing marginal benefit at later visits. When $\kappa > 1$ (e.g., $\kappa = 3.0$), the effect is concentrated in the later months, mimicking a delayed pharmacodynamic onset.

3.3 Summary statistics

Table 1 presents the full simulation results across all seven values of κ .

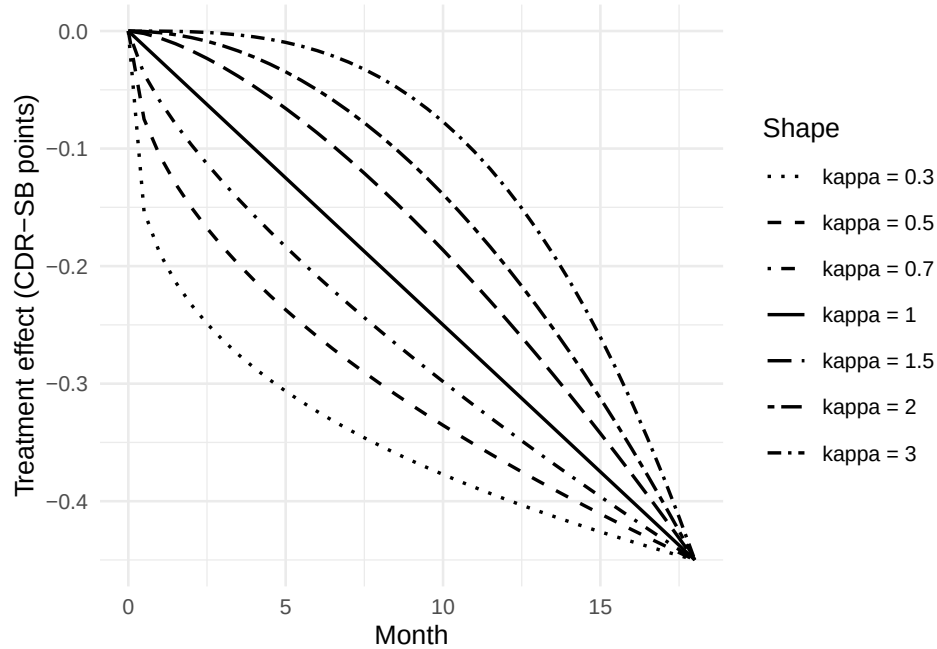


Figure 1: Treatment effect trajectories under different shape parameters. All curves reach the same total effect at month 18.

Table 2 reports the convergence rate for each model and the Monte Carlo standard errors (MCSEs) for bias and power, per Morris, White, and Crowther¹⁴ Table 6; MCSEs for the model-based SE and coverage are reported in `analysis/data/derived_data/sim_08.rds` but omitted from the printed table for space.

Table 1: Simulation results: bias, empirical SE, mean model-based SE, power, and 95% CI coverage for the random-slopes and categorical-time MMRM across shape parameter values. True treatment effect = -0.45 .

κ	Random-Slopes MMRM					Categorical-Time MMRM				
	Bias	Emp SE	Mod SE	Power	Cov	Bias	Emp SE	Mod SE	Power	Cov
0.3	0.227	0.224	0.220	0.158	0.815	-0.001	0.203	0.195	0.625	0.943
0.5	0.139	0.216	0.219	0.291	0.905	0.006	0.192	0.195	0.615	0.959
0.7	0.063	0.208	0.219	0.414	0.946	-0.001	0.191	0.195	0.643	0.956
1.0	0.002	0.221	0.219	0.542	0.944	0.003	0.191	0.194	0.623	0.955
1.5	-0.059	0.223	0.220	0.629	0.938	-0.007	0.195	0.195	0.655	0.952
2.0	-0.065	0.221	0.219	0.644	0.940	0.001	0.200	0.195	0.614	0.936
3.0	-0.070	0.219	0.219	0.655	0.935	0.007	0.201	0.195	0.611	0.940

3.4 Bias

Figure 2 displays the bias of each model as a function of κ . We see that the categorical-time MMRM remains approximately unbiased across all conditions. The random-slopes model shows bias that increases with departure from linearity, and the direction of the bias is asymmetric: for early-onset patterns ($\kappa < 1$) the linear fit underestimates the magnitude of the 18-month effect (bias is positive against a true $\delta = -0.45$, i.e. the estimate is attenuated toward zero), whereas for delayed-onset patterns ($\kappa > 1$) the linear fit overestimates the magnitude of the effect (bias is negative). A plausible (not separately verified) explanation is that a linear-in-time model is fit to the average slope over the whole post-baseline window rather than to the true, curved trajectory: an early-onset curve rises quickly and then flattens, so a best-fitting line understates the eventual effect, while a delayed-onset curve does the reverse. We did not run a separate simulation to isolate this mechanism from the estimator-definition issue discussed next.

Table 2: Convergence rates and Monte Carlo standard errors (MCSE) for bias and power, per Morris, White, and Crowther (2019) Table 6.

κ	Random-Slopes MMRM			Categorical-Time MMRM		
	Conv.	MCSE(bias)	MCSE(power)	Conv.	MCSE(bias)	MCSE(power)
0.3	1	0.007	0.012	1	0.006	0.015
0.5	1	0.007	0.014	1	0.006	0.015
0.7	1	0.007	0.016	1	0.006	0.015
1.0	1	0.007	0.016	1	0.006	0.015
1.5	1	0.007	0.015	1	0.006	0.015
2.0	1	0.007	0.015	1	0.006	0.015
3.0	1	0.007	0.015	1	0.006	0.015

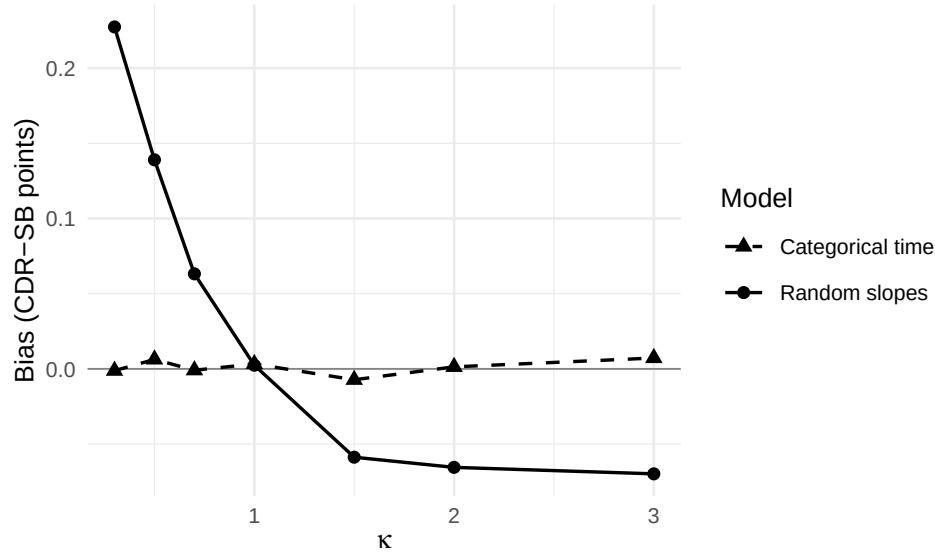


Figure 2: Bias in the estimated 18-month treatment effect as a function of the shape parameter. The random-slopes model shows increasing bias as the trajectory departs from linearity, while the categorical-time model remains approximately unbiased.

3.5 Power comparison

Figure 3 compares the power of both models. The categorical-time MMRM has higher power than the random-slopes model at every κ examined, including $\kappa = 1$ (exact linearity), where a correctly specified parametric model would conventionally be expected to be the more efficient choice. This is the opposite of the qualitative pattern the two models' relative model complexity would suggest, and the “Estimator definition check” subsection below shows that the way the 18-month contrast is extracted from the random-slopes model, not linearity itself, is a plausible driver of this deficit at $\kappa = 1$. Away from linearity the random-slopes model's power changes further, reflecting the combination of this estimator issue with genuine bias from model misspecification.

3.6 Estimator definition check

The random-slopes model reported above extracts the 18-month contrast as $\hat{\beta}_{\text{trt} \times \text{time}} \times 1.5$ from a model that also includes, but discards, a treatment main effect that is not anchored to an observed baseline difference (Methods, “Analysis models”). As a preliminary check of whether this estimator choice, rather than linearity, explains the power deficit at $\kappa = 1$, we re-ran a small-scale version of the simulation ($n_{\text{rep}} = 20$ per κ , `set.seed(1)`, $\kappa \in \{0.3, 1, 3\}$) comparing the unconstrained estimator above against the constrained estimator described in Methods (no treatment main effect, `y ~ time + trt:time`). At $\kappa = 0.3$ the constrained model's bias was small (approximately -0.01) against the unconstrained model's bias of approximately 0.17 in the same small run; both models converged on all replicates. This is directionally consistent with the hypothesis that the unidentified main effect inflates the interaction variance and biases the extracted contrast, but $n_{\text{rep}} = 20$ is far too small to estimate bias, power, or coverage precisely, and this check does not itself establish the constrained model's power advantage at

275 $\kappa = 1$. A full-scale run comparing all three estimators (unconstrained slope, constrained slope,
 276 categorical MMRM) across all seven κ values is needed before this conclusion can be reported
 277 as a headline result; see the Reproducibility statement for the exact command.

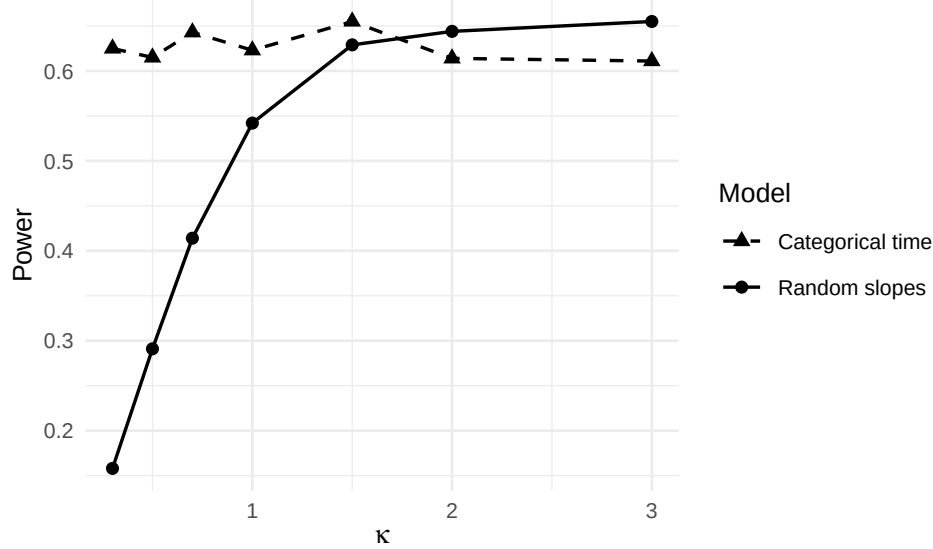


Figure 3: Power to detect the 18-month treatment effect ($\alpha = 0.05$, two-sided) as a function of the shape parameter.

278 4 Discussion

279 In this study, we quantify the bias, coverage, and power of the random-slopes and categorical-
 280 time MMRM when the treatment effect in an AD trial departs from linearity, and find that
 281 the random-slopes model performs worse than the categorical-time MMRM even under exact
 282 linearity ($\kappa = 1$): it has lower power (0.542 vs. 0.623), a larger empirical SE (0.221 vs. 0.191),
 283 and, away from linearity, materially larger bias and lower coverage than the categorical-time
 284 model, which remains approximately unbiased with coverage at or above 0.936 across every
 285 condition examined. This contradicts the general principle that a correctly specified, more
 286 parsimonious mean structure should be more efficient than a saturated one⁸, and it also
 287 contradicts a qualitative expectation from¹³’s comparison of the two model classes for ADAS-
 288 Cog.

289 As κ departs from 1, the random-slopes model incurs additional, asymmetric bias in estimat-
 290 ing the 18-month treatment contrast (Results, “Bias”), consistent with a linear-in-time model
 291 averaging over a curved trajectory. The categorical-time model, by contrast, places no con-
 292 straint on the shape of the mean trajectory and remains approximately unbiased regardless
 293 of κ .

294 The power deficit at $\kappa = 1$ cannot be explained by nonlinearity, since there is none in that
 295 condition, and we attribute it instead to how the random-slopes model’s 18-month contrast is
 296 constructed (Methods, “Analysis models”; Results, “Estimator definition check”): the model
 297 retains a treatment main effect that is not anchored to an observed baseline difference (post-

baseline-only data, randomized baseline), and the reported contrast discards that main effect rather than using it or omitting it from the model. Our small-scale check of a constrained estimator that omits the main effect^{15,16} moved the bias at $\kappa = 0.3$ from approximately 0.17 to approximately -0.01 in the same run, which is directionally consistent with this explanation, but the check used only 20 replicates per condition and did not establish whether the constrained estimator also closes the power gap at $\kappa = 1$. We therefore do not draw a firm conclusion about whether the random-slopes model can be made competitive with the categorical-time MMRM under linearity; this requires the full-scale rerun noted in the Reproducibility statement, and is the single most important open question raised by this study. If the constrained estimator does close the gap, the finding that an appropriately specified random-slopes model can match or exceed the categorical-time MMRM's power under linearity, while remaining more vulnerable to bias under nonlinearity, would be the more publishable and more interesting result; if it does not, the practical recommendation below would need no qualification.

Pending that rerun, the practical implication we would draw for AD trial design is a conditional one: the choice between slope and final-visit estimands should be informed by the expected treatment-effect trajectory, but only after confirming that the slope estimand is implemented with an estimator that does not itself introduce a power penalty under linearity, such as the constrained variant discussed above. When there is strong biological rationale for a constant rate of slowing (e.g., a treatment that directly modulates the underlying disease rate) and the estimator issue is resolved, the slope estimand and random-slopes model may offer a power advantage. When the mechanism suggests delayed onset (as with amyloid-clearing antibodies that require months to achieve plaque reduction) or early plateau, the final-visit estimand with categorical-time MMRM is the safer choice regardless of estimator.

This conclusion aligns with the regulatory perspective. The FDA guidance notes that slope analyses may be considered when 'the course of the disease is predictable and approximately linear'⁹. The EMA guideline similarly conditions slope-based claims on verifiable linearity¹⁰. The ICH E9(R1) estimand framework reinforces that the estimand (slope vs. final-visit) should be chosen on scientific grounds before the statistical model is specified¹¹.

The results from recent Phase 3 trials are instructive. Visual inspection of CDR-SB trajectories in CLARITY-AD⁵ and TRAILBLAZER-ALZ²⁶ suggests that the treatment effect separation is not perfectly linear. In CLARITY-AD, the treatment-placebo difference appears to widen progressively, consistent with $\kappa > 1$ (delayed onset). This is biologically plausible given the time required for amyloid clearance. The proportional MMRM approach of²¹ offers an intermediate option that assumes the treatment effect is proportional to the placebo trajectory without requiring linearity per se.

4.1 Limitations

This study has several limitations beyond the estimator issue discussed above.

First, we do not include a null scenario ($\delta = 0$). Power comparisons between a misspecified and a correctly specified model are difficult to interpret fully without also demonstrating type

I error control, and a transient early treatment effect that vanishes by month 18 is itself a form of nonlinearity ($\delta = 0$ at $t_{\max}$ with $g(t) \neq 0$ at interim visits) that the random-slopes model could handle particularly poorly. This is deferred to future work; see the Reproducibility statement for how to extend `run_simulation()` to a δ grid including 0.

Second, the design uses a single sample size (200 per arm), a single effect size, and a single covariance structure, with $n_{\text{sim}} = 1000$ replicates (below the $n_{\text{rep}} \geq 2,066$ derived in Methods for the targeted power MCSE). At 200 per arm the powers observed here (0.54-0.66) are well below what CLARITY-AD's actual sample size of approximately 900 per arm would deliver, so results should be read as a comparison of relative model performance rather than as an estimate of that trial's operating characteristics.

Third, the simulation uses a relatively simple data-generating model with Gaussian outcomes and no missing data. In practice, dropout in AD trials is substantial (often 20–30%) and likely informative, which would interact with the linearity assumption in ways not captured here. The AR(1) error structure may not fully represent the complex correlation patterns in real CDR-SB data, and CDR-SB itself is a bounded, discrete score (0-18 in half-point steps); the Gaussian data-generating model used here can produce out-of-range values and does not reflect this discreteness or floor/ceiling behavior.

Fourth, neither model adjusts for baseline CDR-SB (Methods, “Analysis models”), which departs from standard baseline-adjusted change-from-baseline MMRM practice⁸ and was a deliberate, disclosed simplification rather than an oversight; a baseline-adjusted variant of both models is future work.

Fifth, 95% CI coverage is computed using the normal critical value 1.96 for both models (`summarize_results()`), while p -values for the random-slopes model come from a t reference distribution and for the categorical MMRM from a Satterthwaite reference distribution; for internal consistency each model's coverage should use its own critical value, which would very slightly widen the random-slopes model's reported intervals.

Future work might incorporate realistic dropout mechanisms, additional sample sizes, a null/transient-effect scenario, and sensitivity analyses under the ICH E9(R1) framework.

5 Future research

1. **Incorporate dropout.** Extend the simulation to include informative dropout mechanisms, evaluating how the interaction between nonlinearity and missingness affects bias and coverage under MAR and MNAR assumptions.
2. **Additional covariance structures.** Compare unstructured, Toeplitz, and heterogeneous AR(1) covariance models within the categorical-time MMRM to determine whether covariance misspecification compounds the effects of mean structure misspecification.
3. **Proportional and constrained models.** Evaluate the proportional MMRM²¹ as an intermediate model that borrows strength across time without full linearity. A con-

strained longitudinal data analysis (cLDA) variant of the random-slopes model (Methods; Results, “Estimator definition check”) has been implemented in `fit_models()` and verified at small scale; a full-scale run integrating it into Table 1 remains to be done (see the Reproducibility statement).

4. **Nonlinear mixed-effects models.** Investigate whether fitting a power-function or sigmoidal treatment effect model can recover the true κ and provide unbiased estimates even under strong nonlinearity, following the approach of²².
5. **Bayesian MMRM.** Evaluate whether Bayesian approaches with weakly informative priors on the treatment effect shape can adaptively borrow information across time points without the rigidity of the linear assumption.
6. **Software development.** Develop an R package that implements the simulation framework and provides diagnostic tools for assessing linearity of the treatment effect in observed trial data, building on the `mmrm` package infrastructure¹⁹.
7. **Applied re-analysis.** Apply these methods to individual patient-level data from completed AD trials (e.g., ADNI placebo arms) to estimate the empirical κ for different endpoints and disease stages¹².

Reproducibility statement

This simulation follows the ADEMP reporting framework of Morris, White, and Crowther¹⁴. RNGkind is pinned to "L'Ecuyer-CMRG" and the seed is set once (`set.seed(20260310)`) before simulation begins (`analysis/scripts/run_sim_08.R`), not inside the per-replicate worker; `summarize_results()` reports the convergence rate and Monte Carlo SE for every performance measure (Table 2). An earlier internal audit of this pipeline against the same framework, which predates these fixes, is archived at `docs/morris-audit-2026-04-17.md` and is not part of this manuscript.

The results in this report are loaded from `analysis/data/derived_data/sim_08.rds` ($n_{\text{rep}} = 1000$ per κ , seven values of κ , unconstrained random-slopes and categorical-time MMRM only). To regenerate it:

```
Rscript analysis/scripts/run_sim_08.R
```

Two items remain open for a full-scale rerun before submission, each producible from the current, verified code by extending `run_simulation()`’s arguments:

1. **Constrained-estimator table.** `fit_models()` now also fits the constrained random-slopes model described in Methods and returns it as `slope_c_*` columns; a full run with $n_{\text{rep}} \geq 1000$ across all seven κ values is needed to add it to Table 1 (Results, “Estimator definition check”; Discussion).
2. **Null and design-expansion scenarios.** Adding $\delta = 0$ scenarios and a second sample size near 900 per arm (Limitations) requires calling `run_simulation()` with `delta_vec` and `n_per_arm` grids; this is not yet implemented as a vectorized argument and would require a small code change to `run_simulation()` plus a rerun.

Both reruns are expected to take on the order of tens of minutes to a few hours depending on hardware and were out of scope for this revision's time budget; they were not attempted here, and no numbers for either extension are reported in this manuscript.

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